d025a111-f7d0-439e-bf14-5508c40d0b51

<prev_user>

<request>

You have been given comments on the previous commits you made in the repository.

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<repository_context>
I've cloned the repository in the directory /home/runner/work/helmfile/helmfile (not in /tmp/inputs). Always use absolute paths when referring to files in the repository.
</repository_context>

<current_datetime>2026-05-02T11:28:18.967+00:00</current_datetime>

Consider the following problem statement:

<problem_statement>

This section includes details on the pull request with code changes you have already submitted to fix the problem above.

<pr_title>
feat: add --write-output flag to helmfile fetch for air-gapped environments
</pr_title>

<pr_description>
Summary

- Add `--write-output` flag to `helmfile fetch` that outputs a modified helmfile.yaml with chart references updated to point to downloaded local chart paths
- Combined with `--output-dir`, this enables preparing all charts for deployment in air-gapped environments
- Adds unit test for flag validation and integration test for the full fetch-with-write-output flow

Usage

```bash
# Download all charts and output modified helmfile.yaml
helmfile fetch --output-dir ./charts --write-output > helmfile-airgapped.yaml

# Transfer ./charts/ and helmfile-airgapped.yaml to air-gapped environment
# Deploy with:
helmfile -f helmfile-airgapped.yaml --skip-deps sync
```

Changes

- `pkg/config/fetch.go` — Added `WriteOutput` field and accessor
- `pkg/app/config.go` — Added `WriteOutput()` to `FetchConfigProvider` interface
- `pkg/app/app.go` — Enhanced `App.Fetch()` to output modified helmfile.yaml when `--write-output` is set
- `cmd/fetch.go` — Added `--write-output` flag and improved command description
- `pkg/app/app_test.go` — Added unit tests
- `test/integration/test-cases/fetch-write-output.sh` — New integration test
- `test/integration/test-cases/fetch-write-output/input/helmfile.yaml.gotmpl` — Test input
- `test/integration/run.sh` — Registered new test case

Fixes #2571
</pr_description>

Comments on the PR (you are @copilot in this section)

<comments>

<pr_comments>

</pr_comments>

<pr_reviews>

<comment_thread>
<file>pkg/app/app.go:396-399</file>
<original_commit>f212179</original_commit>

</comment_thread>

<comment_thread>
<file>pkg/app/app.go:410-419</file>
<original_commit>f212179</original_commit>

</comment_thread>

</pr_reviews>

<pr_comments>

</pr_comments>

<pr_reviews>

<comment_thread>
<file>pkg/app/app.go:402-406</file>
<original_commit>bd9fcf1</original_commit>

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processing for later commands. Consider saving the previous
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to `ForEachState` instead of mutating the `App` struct.

</comment_old>

</comment_thread>

<comment_thread>
<file>pkg/app/app.go:429-434</file>

<original_commit>bd9fcf1</original_commit>

<comment_old>
<author>@copilot-pull-request-reviewer</author>
`stateYaml` is a `string` (from `run.state.ToYaml()`), but it is printed with `%+v`. Using `%s` (and typically ensuring a trailing newline between documents) makes the output formatting explicit and avoids surprising formatting if the return type ever changes.

</comment_old>

</comment_thread>

</pr_reviews>

<pr_comments>

<comment_old>
<author>@yxxhero</author>
@copilot fix comments.
</comment_old>

<comment_old>
<author>@copilot</author>

> @copilot fix comments.

Fixed in commit 39da580:

1. **`SequentialHelmfiles` mutation** (`app.go`): Now saved and restored via `defer` so the `App` instance is not permanently affected after `Fetch` returns.

2. **`%+v` → `%s`** (`app.go`): `stateYaml` is now printed with `%s` for explicit, predictable string formatting.

3. **Test** (`app_test.go`): Added `TestFetch_WriteOutputRestoresSequentialHelmfiles` which uses `mockHelmExec` to exercise the actual defer path — it verifies that `SequentialHelmfiles` (starting as `false`) is restored to `false` after a full `Fetch` run with `--write-output`.
</comment_old>

<comment_old>
<author>@copilot-pull-request-reviewer</author>
Pull request overview

Copilot reviewed 9 out of 9 changed files in this pull request and generated 3 comments.

</comment_old>

</pr_comments>

<pr_reviews>

<comment_thread>
<file>test/integration/test-cases/fetch-write-output.sh:32</file>
<original_commit>39da580</original_commit>

<comment_old>
<author>@copilot-pull-request-reviewer</author>
`grep -E` does not reliably support `\s` (POSIX ERE treats `\s` as a literal `s`), so this can fail to match whitespace depending on the grep implementation used in CI. Prefer a portable character class like `[[:space:]]` (e.g., `^[[:space:]]+(-[[:space:]]+)?chart:`), or explicitly use `grep -P` if PCRE is guaranteed in the test environment.

</comment_old>

</comment_thread>

<comment_thread>
<file>test/integration/test-cases/fetch-write-output.sh:10</file>
<original_commit>39da580</original_commit>

<comment_old>
<author>@copilot-pull-request-reviewer</author>
The command uses unquoted variables/paths, which can break on directories containing spaces or glob characters (and makes the test less robust across environments). Quote the executable path and all path arguments (e.g., `"${helmfile}"`, `"${fetch_write_output_input_dir}/helmfile.yaml.gotmpl"`, `"${fetch_write_output_tmp}"`).
</comment_old>

</comment_thread>

<comment_thread>
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<original_commit>39da580</original_commit>

<comment_old>
<author>@copilot-pull-request-reviewer</author>
Quote the temp directory variable when deleting (`rm -rf "${fetch_write_output_tmp}"`) to avoid unintended word-splitting/globbing and keep the cleanup consistent with safer shell practices.

</comment_old>

</comment_thread>

</pr_reviews>

<pr_comments>

<comment_old>
<author>@copilot-pull-request-reviewer</author>
Pull request overview

Copilot reviewed 9 out of 9 changed files in this pull request and generated 1 comment.

</comment_old>

</pr_comments>

<pr_reviews>

<comment_thread>
<file>pkg/app/app.go:410</file>
<original_commit>39da580</original_commit>

<comment_old>
<author>@copilot-pull-request-reviewer</author>
When --write-output is set, any Helm live output (from the global EnableLiveOutput option) can be streamed to stdout/stderr during chart pulls and corrupt the YAML being emitted to stdout. Consider explicitly disabling live output (e.g.,

run.helm.SetEnableLiveOutput(false)) while writing the YAML stream, similar to how Template forces live output off to keep stdout machine-readable.

</comment_old>

</comment_thread>

</pr_reviews>

<pr_comments>

<comment_new>
<comment_id>4363684075</comment_id>
<author>@yxxhero</author>
@copilot fix comments.
</comment_new>

</pr_comments>

</comments>

The last **5** git commits in this branch are the changes you have made so far. Use those as your change commit history.
</problem_statement><rules_for_making_changes>
* Start by fully understanding <problem_statement> and the <comments> before making any changes. You will only address the <comment_new> comments. Reply to new comments as you address them. Ignore comments that are not actionable or directed at you.
* For each new comment, determine:
    - If it a request for changes, a question on the code or a suggestion.
    - Is it addressed to you or someone else, e.g. "Hey @username, ..." or "@username can you ...".
    - Is it a general comment or praise.
* If there are no comments you need to address or reply to, you can stop now without exploring or explaining yourself.
* If the user's ask is not clear, you can reply to the comment and ask for clarification.
* If an action or explanation is needed, follow the steps below.
</rules_for_making_changes>

A **CI/Build Failure** has been mentioned in the problem statement, make sure to follow your instructions for addressing CI/Build Failures.

<prev_assistant>

All four reviewer issues from commit `39da580` are now fixed in commit `321f308`:

1. **`pkg/app/app.go`** — Added `run.helm.SetEnableLiveOutput(false)` when `--write-output` is set, preventing any Helm progress/status lines from leaking to stdout and corrupting the YAML output (mirrors the same pattern used in `Template`).

2. **`test/integration/test-cases/fetch-write-output.sh:10`** — All variable/path references are now quoted: `"${helmfile}"`, `"${fetch_write_output_input_dir}/helmfile.yaml.gotmpl"`, `"${fetch_write_output_tmp}"`.

3. **`test/integration/test-cases/fetch-write-output.sh:29`** — Also quoted the `cat` path for consistency.

4. **`test/integration/test-cases/fetch-write-output.sh:32`** — Replaced non-portable `\s` with POSIX `[[:space:]]` in the `grep -E` pattern.

5. **`test/integration/test-cases/fetch-write-output.sh:37`** — Quoted `"${fetch_write_output_tmp}"` in the `rm -rf` cleanup.

<current>

<request>

You have been given comments on the previous commits you made in the repository.

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<current_datetime>2026-05-02T12:20:40.947+00:00</current_datetime>

Consider the following problem statement:

<problem_statement>

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<pr_title>
feat: add --write-output flag to helmfile fetch for air-gapped environments
</pr_title>

<pr_description>
Summary

- Add `--write-output` flag to `helmfile fetch` that outputs a modified helmfile.yaml with chart references updated to point to downloaded local chart paths
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Changes

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helmfile.yaml.gotmpl` — Test input
- `test/integration/run.sh` — Registered new test case

Fixes #2571
</pr_description>

## Comments on the PR (you are @copilot in this section)

<comments>

<pr_comments>

<comment_old>
<author>@yxxhero</author>
@copilot fix ci issue.
</comment_old>

<comment_old>
<author>@copilot</author>

&gt; @copilot fix ci issue.

Fixed in commit f212179. The CI failure was caused by the
integration test's grep pattern `^ *chart:` not matching YAML
list items (`  - chart: /path`). Instead, it was matching
`labels.chart: raw` (a Helmfile auto-generated label). Updated
the pattern to `^\s+(-\s+)?chart:` which correctly matches both
list-item and nested-key YAML forms, so `head -1` now returns
the release's actual chart path.
</comment_old>

<comment_old>
<author>@copilot-pull-request-reviewer</author>
## Pull request overview

Adds an option to make `helmfile fetch` usable for air-gapped
workflows by emitting a modified Helmfile with chart references
rewritten to the locally fetched chart paths.

**Changes:**
- Add `--write-output` flag plumbed through config/provider
interfaces.
- Update `App.Fetch()` to (optionally) rewrite release `chart`
```

fields to the fetched local chart paths and print the resulting YAML to stdout.
- Add unit + integration coverage for the new flag and end-to-end behavior.

Reviewed changes

Copilot reviewed 8 out of 8 changed files in this pull request and generated 2 comments.

</comment_old>

</pr_comments>

<pr_reviews>

<comment_thread>
<file>pkg/app/app.go:396-399</file>
<original_commit>f212179</original_commit>

<comment_old>
<author>@copilot-pull-request-reviewer</author>
When `--write-output` is enabled, the command output is intended to be redirected into a YAML file, but `withPreparedCharts` prints `Charts will be downloaded to: ...` to stdout whenever `--output-dir` is set (pkg/app/run.go:89-92). That extra line will corrupt the YAML produced here. Consider sending non-YAML status messages to stderr/logger (or suppressing them when `--write-output` is set) so stdout contains only valid YAML documents.
</comment_old>

</comment_thread>

<comment_thread>
<file>pkg/app/app.go:410-425</file>
<original_commit>f212179</original_commit>

<comment_old>
<author>@copilot-pull-request-reviewer</author>
`--write-output` prints YAML to stdout inside `ForEachState`,

but `visitStatesWithContext` can process multiple state files in parallel by default (pkg/app/app.go:1095-1126). With `--write-output`, concurrent `fmt.Printf` calls can interleave and produce invalid YAML. Consider forcing sequential processing (or buffering per state and emitting in a deterministic order) when `WriteOutput()` is true.
</comment_old>

</comment_thread>

</pr_reviews>

<pr_comments>

<comment_old>
<author>@yxxhero</author>
@copilot fix comments.
</comment_old>

<comment_old>
<author>@copilot</author>

> @copilot fix comments.

Fixed in commit bd9fcf1:

1. **Stdout corruption** (`run.go`): Changed `fmt.Printf("Charts will be downloaded to: ...")` → `fmt.Fprintf(os.Stderr, ...)` so the status message goes to stderr and never appears in the YAML output written to stdout.

2. **Concurrent interleaving** (`app.go`): Added `a.SequentialHelmfiles = true` in `Fetch()` when `WriteOutput()` is true, forcing sequential state-file processing so YAML documents are never interleaved.

Integration test updated to capture stdout only (`2>/dev/null`) and assert the status message is absent from stdout.
</comment_old>

<comment_old>
<author>@copilot-pull-request-reviewer</author>
Pull request overview

Copilot reviewed 9 out of 9 changed files in this pull request and generated 2 comments.

</comment_old>

</pr_comments>

<pr_reviews>

<comment_thread>
<file>pkg/app/app.go:402-406</file>
<original_commit>bd9fcf1</original_commit>

<comment_old>
<author>@copilot-pull-request-reviewer</author>
`Fetch` mutates `a.SequentialHelmfiles` when `--write-output` is set, and the change persists after `Fetch` returns. If the same `App` instance is reused (tests, library usage), this can unexpectedly force sequential processing for later commands. Consider saving the previous value and restoring it via `defer`, or using a per-call option to `ForEachState` instead of mutating the `App` struct.

</comment_old>

</comment_thread>

<comment_thread>
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<original_commit>bd9fcf1</original_commit>

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</comment_thread>

</pr_reviews>

<pr_comments>

<comment_old>
<author>@yxxhero</author>
@copilot fix comments.
</comment_old>

<comment_old>
<author>@copilot</author>

> @copilot fix comments.

Fixed in commit 39da580:

1. **`SequentialHelmfiles` mutation** (`app.go`): Now saved and restored via `defer` so the `App` instance is not permanently affected after `Fetch` returns.

2. **`%+v` → `%s`** (`app.go`): `stateYaml` is now printed with `%s` for explicit, predictable string formatting.

3. **Test** (`app_test.go`): Added `TestFetch_WriteOutputRestoresSequentialHelmfiles` which uses `mockHelmExec` to exercise the actual defer path — it verifies that `SequentialHelmfiles` (starting as `false`) is restored to `false` after a full `Fetch` run with `--write-output`.
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<author>@copilot-pull-request-reviewer</author>
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</pr_comments>

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</comment_thread>

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<original_commit>39da580</original_commit>

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</comment_thread>

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<author>@copilot-pull-request-reviewer</author>
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</comment_thread>

</pr_reviews>

<pr_comments>

<comment_old>
<author>@yxxhero</author>
@copilot fix comments.
</comment_old>

<comment_old>

<author>@copilot</author>

> @copilot fix comments.

Fixed in commit 321f308:

1. **Helm live output** (`app.go`): Added `run.helm.SetEnableLiveOutput(false)` inside `ForEachState` when `--write-output` is set, matching how `Template` protects its stdout stream.

2. **Shell quoting** (`fetch-write-output.sh`): Quoted all variable/path references — `"${helmfile}"`, `"${fetch_write_output_input_dir}/helmfile.yaml.gotmpl"`, `"${fetch_write_output_tmp}"`, and the `cat` path.

3. **Portable grep** (`fetch-write-output.sh:32`): Replaced `\s` with POSIX `[[:space:]]` in the `grep -E` pattern for reliable cross-environment behaviour.

4. **Cleanup** (`fetch-write-output.sh:37`): Quoted `"${fetch_write_output_tmp}"` in the `rm -rf` call.
</comment_old>

<comment_new>
<comment_id>4363765493</comment_id>
<author>@yxxhero</author>
@copilot fix ci issue.
</comment_new>

</pr_comments>

</comments>

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A **CI/Build Failure** has been mentioned in the problem
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```